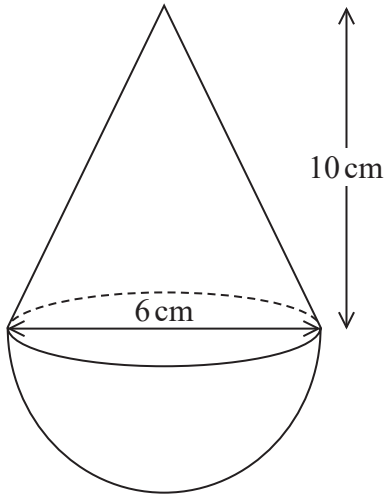
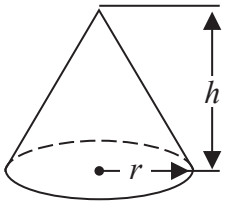
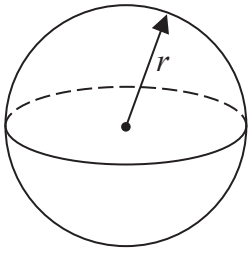
15 The diagram shows a solid shape.
The shape is a cone on top of a hemisphere.



Volume of a cone = $\frac{1}{3}\pi r^2 h$



Volume of a sphere = $\frac{4}{3}\pi r^3$



The height of the cone is 10 cm.
The base of the cone has a diameter of 6 cm.
The hemisphere has a diameter of 6 cm.

The total volume of the shape is $k\pi$ cm^3, where k is an integer.

Work out the value of k.

Both radii are 6 ÷ 2 = 3 cm.

Cone: (1/3) × π × 3^2 × 10 = 30π cm^3

1 mark
Use a correct volume formula.

Hemisphere: (1/2)(4/3)π × 3^3 = 18π cm^3

1 mark
Include the hemisphere.

Total = 30π + 18π = 48π cm^3

1 mark
Simplify at least one volume.

1 mark
k = 48.

$k =$ 48

(Total for Question 15 is 4 marks)